14/01/2026

Code-C

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720 NCERT Booster Test Series for NEET-2026_RM(P3)_NBTS-02C **Time : 180 Min.**

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves, Electric Charges & Fields, Electrostatic Potential and Capacitance, Current Electricity
Chemistry: Equilibrium, Redox Reactions, Electrochemistry, Solutions, Chemical Kinetics, Organic Chemistry: Some Basic Principle & Techniques, Hydrocarbons, Haloalkanes and Haloarenes
Botany: Anatomy of Flowering Plants, Plant Kingdom, Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants
Zoology: Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration, Animal Kingdom, Structural Organisation in Animals–Animal Morphology (Cockroach & Frog)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

Vivek Krishna

PHYSICS

1. A particle of mass M is situated at the centre of spherical shell of mass M and radius a. The gravitational potential at the surface of the shell will be

(1) $\frac{-GM}{a}$

(2) $\frac{-2GM}{a}$

(3) $\frac{-GM}{2a}$

(4) $\frac{-GM}{4a}$

2. If force of gravitation varies as $F \propto r^2$, then the orbital speed of satellite v_0 will be proportional to (where r is radius of orbit)

(1) $v_0 \propto r^{\frac{-1}{2}}$

(2) $v_0 \propto r^2$

(3) $v_0 \propto r^{3/2}$

(4) $v_0 \propto r$

3. A : Escape velocity is independent of the mass of the body and does not depend on direction of projection of the body.
R : If velocity of projection is less than escape velocity, then it will reach a certain height and then may either move in an orbit around the planet or may fall back on the planet.

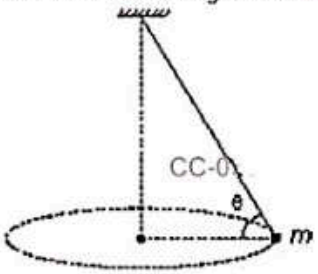
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

4. A stone of mass m tied to one end of an elastic thread of length l. The diameter of the thread is d and it is suspended vertically. The stone is now rotated in a horizontal plane and makes an angle θ with horizontal. If Young's modulus is Y then increase in length of thread is



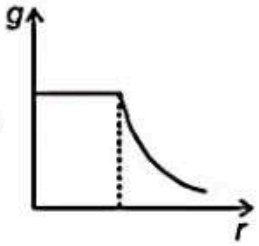
(1) $\frac{mgl}{\pi d^2 Y \sin\theta}$

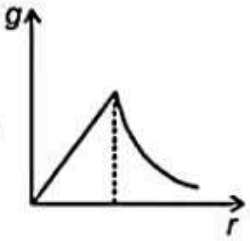
(2) $\frac{4mgl}{\pi d^2 Y \cos\theta}$

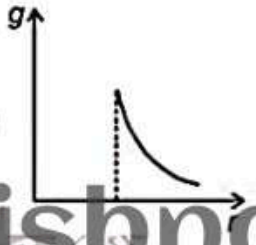
(3) $\frac{4mgl}{\pi d^2 Y \sin\theta}$

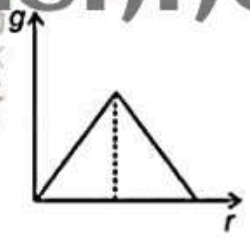
(4) $\frac{mgl}{\pi d^2 Y \cos\theta}$

5. Which of the following curves show the variation of acceleration due to gravity from the centre of earth to the points far away from the surface of earth?

(1)


(2)


(3)


(4)


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6. Two wires of same diameter of the same material having the length $2L$ and $3L$. If the force F is applied on each, the ratio of the work done on the two wires will be

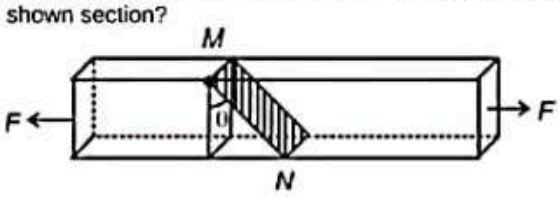
(1) 1 : 3

(2) 1 : 4

(3) 2 : 1

(4) 2 : 3

7. A rod, shown in figure, is subjected to equal and opposite forces. *MN* is a plane making an angle θ = 45° with cross-section *A* of the rod. What is tensile stress on *MN* shaded shown section?



(1) $\frac{\sqrt{2}F}{A}$

(2) $\frac{F}{2A}$

(3) $\frac{F}{A}$

(4) $\frac{2F}{A}$

8. A cube of density 250 kg/m^3 floats in water then what part of total volume of the cube outside the water?

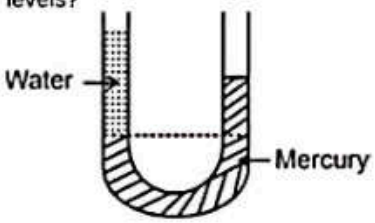
(1) 75%

(2) 25%

(3) 33.3%

(4) 67.7%

9. A *U* tube open at both ends contains mercury. 15 cm of water is added to left arm. What is difference of height of two levels?



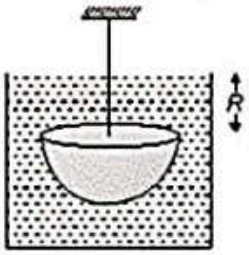
(1) 7 cm

(2) 13.9 cm

(3) 8 cm

(4) 4.1 cm

10. A uniform solid hemisphere of radius *R* is immersed into a liquid of density ρ as shown. The base of the hemisphere is at distance *R* from the surface of the liquid. Ignore atmospheric pressure. The force by the liquid on the curved surface of the hemisphere is



(1) $\pi\rho g R^3$

(2) $\frac{2\pi}{3}\rho g R^3$

(3) $\frac{5\pi}{3}\rho g R^3$

(4) $\frac{4\pi}{3}\rho g R^3$

11. The pressure of water in a water pipe when the tap is closed and opened are respectively 2.5 × 10^5 N/m^2 and 2 × 10^5 N/m^2. The velocity of water flowing through the pipe when tap is opened is

(1) 10 m/s

(2) 20 m/s

(3) 5 m/s

(4) 15 m/s

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12. A water drop has been trapped inside a vertical glass capillary. Select the most appropriate option, regarding shape of trapped water drop.

(1) 

(2) 

(3) 

(4) 

13. 1 kg of ice at 0°C is mixed with 4 kg of water at 30°C. The temperature of mixture is

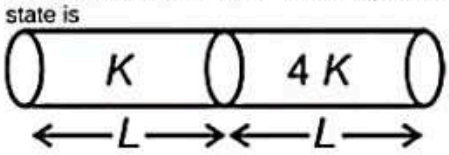
(1) 0°C

(2) 8°C

(3) 20°C

(4) 25°C

14. Two identical cylindrical rods of thermal conductivity K and $4K$ are joined together as shown in figure. The effective thermal conductivities of the composite system in steady state is



(1) $\frac{8K}{5}$

(2) $\frac{2K}{5}$

(3) $\frac{4K}{5}$

(4) $\frac{K}{5}$

15. A sphere, cube and thin circular plate all made of same material and having same mass are heated to a temperature of 300 °C. Which of these objects will cool fastest and which will cool slowest when left in air at room temperature?

(1) Sphere, cube

(2) Cube, circular plate

(3) Circular plate, sphere

(4) All will cool at same rate as mass is same

16. In accordance with Kirchhoff's law for a black body radiation, which one is wrong?

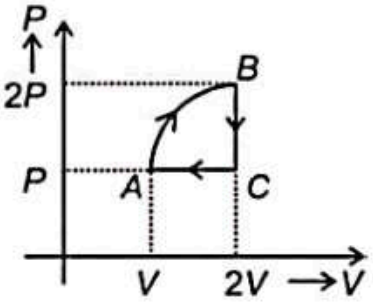
(1) Bad absorber is a bad emitter

(2) Bad reflector is a good emitter

(3) Bad emitter is a good absorber

(4) Bad reflector is a good absorber

17. The pressure-volume (P-V) diagram for the cyclic process ABC is shown in the figure. The work done by the gas in the whole cycle is (part AB of loop is circular)



(1) $\frac{\pi}{4}V^2$

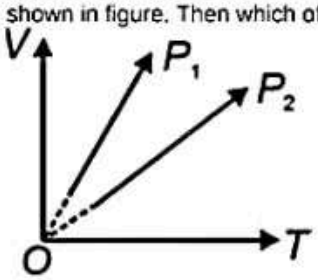
(2) $\frac{\pi}{4}P^2$

(3) $\frac{\pi}{4}PV$

(4) All of these

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18. A monoatomic gas is heated at constant pressure P_1 and P_2. The variation in volume as a function of temperature are shown in figure. Then which of the following is true?



(1) $P_1 = P_2$

(2) $P_1 < P_2$

(3) $P_1 > P_2$

(4) $P_1 \geq P_2$

19. The temperature of hot reservoir of a Carnot's engine is 1500 K. The temperature of sink required to produce 80% efficiency to the engine is

(1) 1200 K

(2) 700 K

(3) 500 K

(4) 300 K

20. When a diatomic gas is heated at constant pressure, the fraction of the heat energy supplied which does work on the surrounding is

(1) $\frac{3}{7}$

(2) $\frac{2}{7}$

(3) $\frac{2}{5}$

(4) $\frac{3}{5}$

21. Three moles of monoatomic ideal gas is mixed with four moles of ideal non linear polyatomic gas ratio of specific heats at constant pressure and constant volume of the gas mixture will be

(1) $\frac{20}{19}$

(2) $\frac{42}{27}$

(3) $\frac{41}{29}$

(4) $\frac{47}{33}$

22. The graph between mean free path and number density of the molecule is look like

(1) Rectangular hyperbola

(2) Parabola

(3) Straight line

(4) Circular

23. Match the columns and tick the correct option

Column I		Column II
a.	Average speed of gas molecules	(i) $\frac{3}{2}k_BT$
b.	Root mean square speed of gas molecules	(ii) $\frac{3}{2}RT$
c.	Average translational kinetic energy of a molecule	(iii) $\sqrt{\frac{3RT}{M}}$
d.	Average translational kinetic energy of 1 mole of gas	(iv) $\sqrt{\frac{8RT}{\pi M}}$

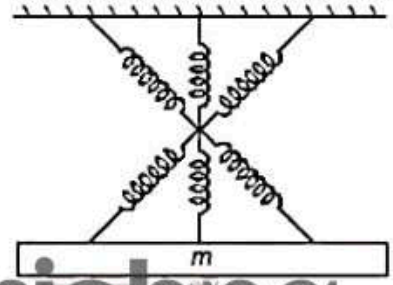
(1) a(iv), b(ii), c(i), d(iii)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iv), b(i), c(iii), d(ii)

24. Six springs each of spring-constant k constitute a simple harmonic motion. Natural frequency of the system is



(1) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{5k}{3m}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{k}{2m}}$

(4) $\frac{1}{2\pi}\sqrt{\frac{3k}{2m}}$

25. In case of forced oscillation of a body

(1) Driving force is constant throughout

(2) Driving force is to be applied only momentarily

(3) Driving force is not required

(4) Driving force has to be periodic and continuous

26. A simple harmonic oscillator has time period T and energy E. The amplitude of oscillator is doubled. Choose the correct option

(1) Time period and energy get doubled

(2) Energy gets doubled but time period remains same

(3) Time period remains same but energy becomes four times

(4) Energy remains same but time period becomes double

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27. The amplitude (A) of damped oscillator becomes half in 5 minutes. The amplitude after next 10 minutes will be

(1) A

(2) $\frac{A}{8}$

(3) $\frac{A}{4}$

(4) $4A$

28. The phenomenon of sound propagation in air is

(1) Isothermal process

(2) Isobaric process

(3) Adiabatic process

(4) Isochoric process

29. A steel wire has a length of 4 m and mass of 6 kg. The speed of transverse wave on this wire under the tension of 6×10^4 N is

(1) 180 ms^{-1}

(2) 200 ms^{-1}

(3) 220 ms^{-1}

(4) 250 ms^{-1}

30. Velocity of sound in air is 320 m/s. For an organ pipe closed at one end of length 2 m, the resonating frequency may be

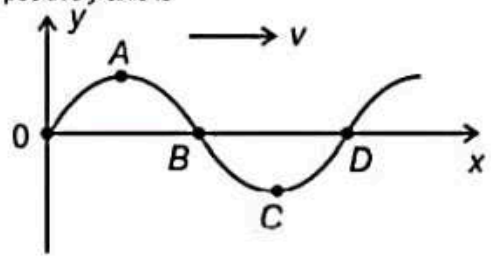
(1) 80 Hz

(2) 160 Hz

(3) 240 Hz

(4) 360 Hz

31. A transverse wave travelling along positive x-direction on a string is shown at an instant. The particle having velocity in positive y axis is



(1) A

(2) B

(3) C

(4) D

32. A tuning fork of unknown frequency produces 6 beats/s when sounded with fork of frequency 510 Hz. The same tuning fork, after waxing, when sounded with fork of frequency 510Hz produces 2 beats/s. Original unknown frequency of the tuning fork will be

(1) 504 Hz

(2) 510 Hz

(3) 516 Hz

(4) 500 Hz

33. A charged particle of charge q and mass m is released from rest in an uniform electric field E. Neglecting the effect of gravity, the kinetic energy of the charged particle after time 't' seconds is

(1) $\frac{qEm}{t}$

(2) $\frac{E^2q^2t^2}{2m}$

(3) $\frac{2E^2t^2}{mq}$

(4) $\frac{Eq^2m}{2t^2}$

34. A point charge 'q' is kept at corner of a cube, then total electric field lines passing through the surface containing the corner having charge, is

(1) Zero

(2) $\frac{q}{12\varepsilon_0}$

(3) $\frac{q}{96\varepsilon_0}$

(4) $\frac{q}{24\varepsilon_0}$

35. The ratio of electric field in axial position to the equatorial position at same distance due to a short dipole is

(1) 2

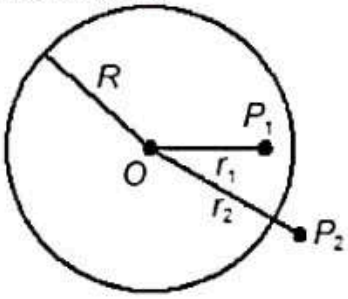
(2) $\frac{1}{2}$

(3) 1

(4) $\frac{1}{4}$

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36. There are two points P_1 and P_2 for a solid non-conducting uniformly charged sphere of radius "R" as shown in figure. If the electric field intensity at P_1 and P_2 are equal in magnitude, then



(1) $R = \sqrt{r_1 r_2}$

(2) $R = r_1^{\frac{2}{5}} r_2^{\frac{4}{5}}$

(3) $R^3 = r_1 r_2^2$

(4) $R = 2\sqrt{r_1 r_2}$

37. If electric potential due to a point charge Q at a distance r from it is taken to be zero, then potential at distance $2r$ from the charge will be

(1) $\frac{kQ}{2r}$

(2) $-\frac{kQ}{2r}$

(3) $-\frac{kQ}{r}$

(4) Zero

38. A parallel plate capacitor (without dielectric) is charged by a battery and kept connected to the battery. A dielectric slab of dielectric constant 'K' is inserted between the plates fully occupying the space between the plates. The energy density of electric field between the plates will

(1) Increase K^2 times

(2) Decrease K^2 times

(3) Increase K times

(4) Decrease K times

39. In a region of space, suppose there exists a electric field $\vec{E} = 40x^3\hat{i}$. The potential difference $V_A - V_O$ is (where V_O is the potential at the origin and V_A the potential at x = 2 m is

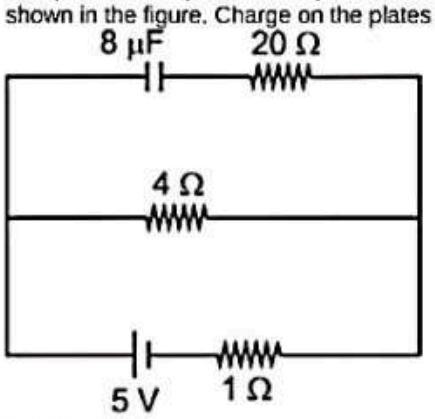
(1) –160 V

(2) 160 V

(3) 120 V

(4) –80 V

40. A capacitor of capacitance 8 μF is connected in a circuit as shown in the figure. Charge on the plates of the capacitor is



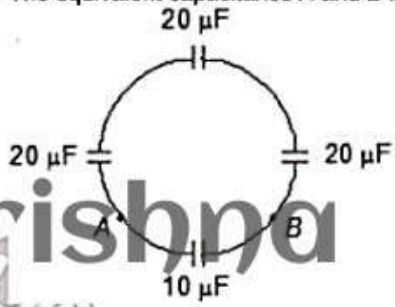
(1) 32 μC

(2) 64 μC

(3) 4 μC

(4) 8 μC

41. The equivalent capacitance A and B is



(1) $\left(\frac{50}{6}\right)$ μF

(2) $\left(\frac{50}{3}\right)$ μF

(3) $\left(\frac{50}{11}\right)$ μF

(4) $\left(\frac{50}{8}\right)$ μF

42. On increasing the temperature of conductor, its resistance increases because the

(1) Relaxation time increases

(2) Mass of electron increases

(3) Electron density decreases

(4) Relaxation time decreases

43. An electric bulb rated as 50 W, 200 V is connected across a 100 V supply. The power consumed by the bulb at this voltage is (Assuming resistance remains same)

(1) 37.5 W

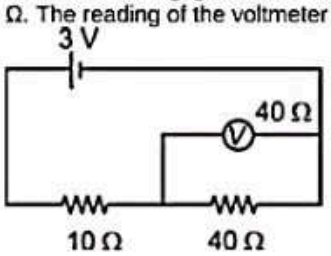
(2) 12.5 W

(3) 25 W

(4) 10 W

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44. As shown in figure, the emf of the cell is 3 V and internal resistance is negligible. The resistance of the voltmeter is 40 Ω. The reading of the voltmeter will be



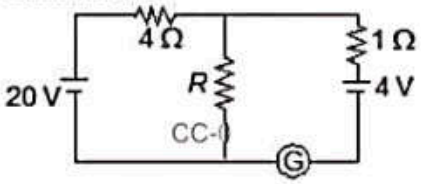
(1) 2 V

(2) 2.4 V

(3) 1 V

(4) 0.6 V

45. If deflection in the galvanometer is zero, then value of resistance *R* is



(1) 3 Ω

(2) 4 Ω

(3) 1 Ω

(4) 2 Ω

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CHEMISTRY

46. If conductivity of 10^{-3} M CH_3COOH solution is 1.85×10^{-5} S cm^{-1} and limiting molar conductivity of H^+ and CH_3COO^- ion are 349.6 S cm^2 mol^{-1} and 40.9 S cm^2 mol^{-1} respectively. The degree of dissociation for 10^{-3} M CH_3COOH will be

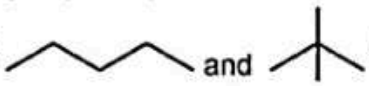
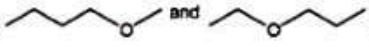
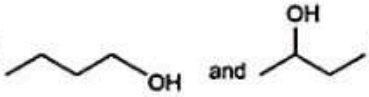
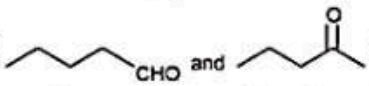
(1) 4.7%

(2) 2.3%

(3) 11.2%

(4) 9.1%

47. Match **List -I** with **List -II**

List -I (Compounds)		**List-II (Isomers)**
(a) 	and	(i) Position isomers
(b) 	and	(ii) Functional group isomers
(c) 	and	(iii) Chain isomers
(d) 	and	(iv) Metamers

Choose the correct answer from the options given below.

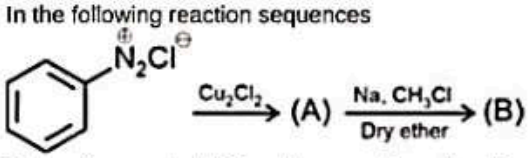
(1) (a)(iv), (b)(iii), (c)(i), (d)(ii)

(2) (a)(i), (b)(iv), (c)(ii), (d)(iii)

(3) (a)(iii), (b)(iv), (c)(ii), (d)(i)

(4) (a)(iii), (b)(iv), (c)(i), (d)(ii)

48. In the following reaction sequences



The major product (B) and name of reaction from A → B formation respectively will be

(1) Toluene and Wurtz-Fittig reaction

(2) Chlorobenzene and Wurtz-Fittig reaction

(3) Toluene and Wurtz reaction

(4) Chlorobenzene and Wurtz reaction

49. Correct expression of K_c for the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ in terms of α is (α is degree of dissociation)

(1) $\frac{\alpha^2}{1-\alpha}$

(2) $\frac{\alpha^2}{4(1-\alpha)}$

(3) $\frac{\alpha^2}{4(1-\alpha)^2}$

(4) $\frac{\alpha}{4(1-\alpha)}$

50. Match batteries in list-I with their electrolyte in list-II.

List-I	List-II
a. Dry cell	(i) 38% (aq) H_2SO_4
b. Mercury cell	(ii) Moist paste of NH_4Cl and $ZnCl_2$
c. $(H_2 - O_2)$ fuel cell	(iii) Paste of KOH and ZnO
d. Lead storage battery	(iv) Conc. (aq) NaOH solution

Correct match is

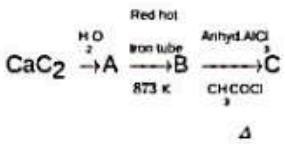
(1) a(ii), b(iii), c(i), d(iv)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(iii), b(ii), c(iv), d(i)

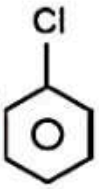
(4) a(ii), b(iii), c(iv), d(i)

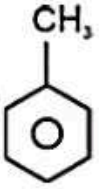
51. Consider the following sequence of reaction.

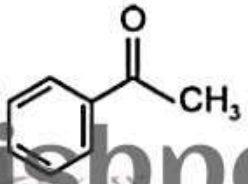


Major product C is

(1) 

(2) 

(3) 

(4) 

52. Given below are two statements.

Statement I: 0.1 m NaCl (aq) has higher boiling point than 0.1 m KCl (aq) if both are completely ionised.

Statement II: Molal elevation constant for (aq) NaCl solution is higher than (aq) KCl solution.

In the light of above statements, choose the **most appropriate** options from the options given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

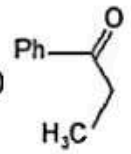
(4) Statement I is incorrect but statement II is correct

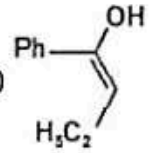
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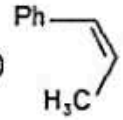
53. In the following reaction

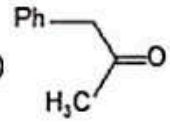
$$Ph - C\equiv C - CH_3 \xrightarrow[H_2O]{Hg^{2+}/H^+} A;$$

'A' is

(1) 

(2) 

(3) 

(4) 

54. The electrode potentials for $M^{2+}(aq) + e^- \rightarrow M^+(aq)$ and $M^+(aq) + e^- \rightarrow M(s)$ are 0.25 V and 0.60 V respectively. The value of $E^*_{M^{2+}/M}$ is

(1) +0.425 V

(2) –0.425 V

(3) +0.855 V

(4) –0.855 V

55. Standard reduction potential for Zn^{2+}/Zn couple is –0.76 V and that for Mg^{2+}/Mg is –2.36 V. These two couples in their standard state are connected to make a cell. The cell potential will be

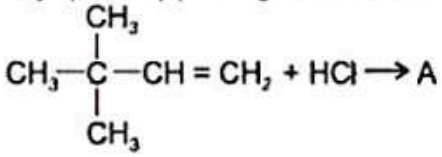
(1) 3.12 V

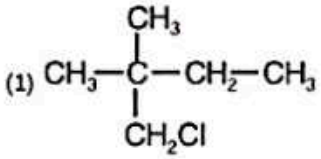
(2) –3.12 V

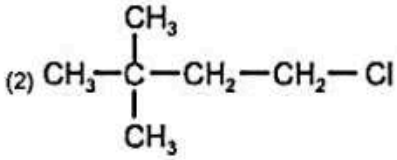
(3) 1.6 V

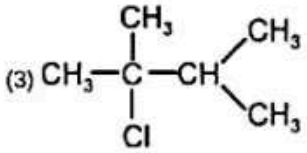
(4) –1.8 V

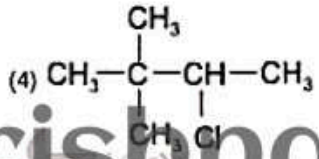
56. Major product (A) of the given reaction is



(1) 

(2) 

(3) 

(4) 

57. Consider the following statements about resonance

(i) Resonating structures are hypothetical and do not represent real molecule.

(ii) The energy of actual structure of the molecule (resonance hybrid) is lower than that of any canonical structures.

(iii) The resonance structures have the same position of nuclei and the same number of unpaired electrons.

The correct statement(s) is/are

(1) (i) and (ii) only

(2) (iii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

58. 3F electricity was passed through molten FeO. Weight of iron metal (Atomic wt = 56) deposited at cathode (in g) is

(1) 112

(2) 84

(3) 56

(4) 28

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68. A mixture of benzene and toluene has 39.52 mm Hg vapour pressure at 300 K. The vapour pressure of pure benzene is 50.71 mm Hg. If mole fraction of toluene in solution is 0.6 then its vapour pressure in pure state will be

(1) 35.06 mm Hg

(2) 38.06 mm Hg

(3) 20.08 mm Hg

(4) 32.06 mm Hg

69. The colligative property widely used to determine molar mass of polymers is

(1) Osmotic pressure

(2) Elevation in boiling point

(3) Depression in freezing point

(4) Relative lowering of vapour pressure

70. Given below are two statements.
Statement (I): In adsorption chromatography, commonly used adsorbents are silica gel and alumina.
Statement (II): Paper chromatography is a type of partition chromatography
In the light of above statements choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

71. Given below are two statements: One is labelled as **Assertion(A)** and the other is labelled as **Reason (R)**.
Assertion(A): Toluene is more reactive than benzene towards electrophilic substitution reaction.
Reason(R): Presence of $-CH_3$ group increases the electron density at o/p positions in toluene and make the benzene ring more reactive towards S_E reaction.
In the light of above statements, select the correct option

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

72. If 100 mL solution of pH = 2 is mixed with 100 mL of solution of pH = 3, then pH of mixture will be

(1) 1.5

(2) 2.26

(3) 0.39

(4) 3.5

73. Match List-I with List-II.

	List-I		List-II van't Hoff factor (i)
a.	Na_2CO_3 (Degree of dissociation 70%)	(i)	3.4
b.	$K_4[Fe(CN)_6]$ (Degree of dissociation 60%)	(ii)	1.8
c.	$Al_2(SO_4)_3$ (Degree of dissociation 75%)	(iii)	2.4
d.	C_6H_5COOH (Degree of dissociation 80%)	(iv)	4

Correct match is

(1) a(i), b(iii), c(iv), d(ii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

74. In the reaction

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(l) + 3H_2O(l)$$

If rate of disappearance of Br^- ion is 4.5×10^{-3} molL^{-1} s^{-1}, then rate of appearance of Br_2 is

(1) 9.0×10^{-4} mol L^{-1} s^{-1}

(2) 1.35×10^{-2} mol L^{-1} s^{-1}

(3) 2.7×10^{-3} mol L^{-1} s^{-1}

(4) 1.5×10^{-3} mol L^{-1} s^{-1}

75. If for any reaction $\ln \frac{k_{300}}{k_{200}} = 7$, where k_{300} and k_{200} are the rate constants at 300 K and 200 K respectively then Activation Energy (E_a) for the reaction is

(1) 1050 R

(2) 2100 R

(3) 4200 R

(4) 8400 R

76. Consider the following statements.
Statement I: Glycerol from spent lye in soap industry is separated by distillation under reduced pressure.
Statement II: Fractional distillation is used to separate different fractions of crude oil in petroleum industry
In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

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77. The following results have been obtained during the kinetic studies of the reaction:
$A + B \rightarrow C$

Experiment	[A]/mol L^{-1}	[B]/mol L^{-1}	Initial rate of formation of C/mol L^{-1} min^{-1}
I	0.1	0.1	1.0×10^{-3}
II	0.3	0.2	4.0×10^{-3}
III	0.3	0.4	1.6×10^{-2}
IV	0.4	0.1	4.0×10^{-3}

The rate law for the reaction will be

(1) $r = K[A]^{-1}[B]^{2}$

(2) $r = K[A]^{1}[B]^{2}$

(3) $r = K[A]^{2}[B]^{1}$

(4) $r = K[A]^{2}[B]^{-1}$

78. On monochlorination of 2-methylbutane the maximum number of optically active isomers formed is

(1) 2

(2) 4

(3) 6

(4) 3

79. If the mass percentage of urea in an aqueous solution is 20% then molality of the solution will be

(1) 5.25 m

(2) 2.35 m

(3) 4.17 m

(4) 4.31 m

80. If half life of a first order reaction is 25 minutes then the time required for 99% completion of reaction will be

(1) 170.25 min

(2) 150.42 min

(3) 200 min

(4) 166.16 min

81. The oxidation state of Cr in CrO_5 is

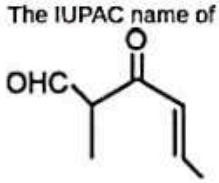
(1) + 10

(2) + 8

(3) + 5

(4) + 6

82. The IUPAC name of compound is



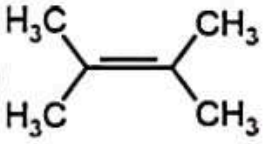
(1) 3-keto-2-methylhex-4-enal

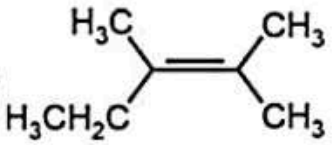
(2) 5-formylhex-2-en-3-one

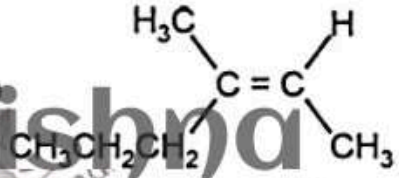
(3) 5-methyl-4-oxohex-2-en-5-al

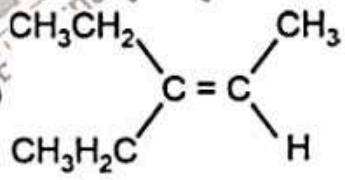
(4) 3-keto-2-methylhex-5-enal

83. An alkene (A) which on reductive ozonolysis yields acetone and butanone. Alkene (A) is

(1) 

(2) 

(3) 

(4) 

84. Which of the following will form a buffer solution?

(1) 100 mL 0.1 M CH_3COOH + 100 mL 0.1 M NaOH

(2) 100 mL 0.1 M CH_3COOH + 50 mL 0.1 M NaOH

(3) 100 mL 0.1 M CH_3COOH + 150 mL 0.1 M NaOH

(4) 100 mL 0.1 M CH_3COOH + 200 mL 0.1 M NaOH

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59. In the following reaction, $C_6H_5Br \xrightarrow[ii)\ C_2H_5OH]{i)\ Mg,\ ether} A$, the product (A) is

(1) C_6H_6

(2) $C_6H_5-C_2H_5$

(3) C_6H_5-OH

(4) $C_6H_5OC_2H_5$

60. Consider the following reaction
$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g),\ \Delta H = -92.2\ kJ$
Choose the condition(s) from the following which favour(s) the formation of methanol.
(a) Decreasing the pressure.
(b) Addition of inert gas at constant volume
(c) Decreasing the temperature

(1) (a) only

(2) (a) and (b) only

(3) (c) only

(4) (a) and (c) only

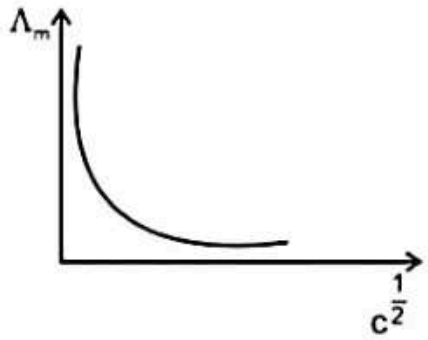
61. Reaction(s) in which H_2O_2 works as an oxidising agent is/are
(a) $H_2O_2 + I_2 \rightarrow H_2O + O_2 + HI$
(b) $H_2O_2 + Ag_2O \rightarrow 2Ag + H_2O + O_2$
(c) $H_2O_2 + PbS \rightarrow PbSO_4 + H_2O$
The **correct** option is

(1) (a) and (b) only

(2) (a) and (c) only

(3) (c) only

(4) (a), (b) and (c)

62. 

Which of the following electrolyte shows the above variation of Λ_m vs $C^{\frac{1}{2}}$?

(1) NaCl

(2) $MgCl_2$

(3) $MgSO_4$

(4) CH_3COOH

63. The rate of reaction A $\rightarrow$ B, is 5.0×10^{-2} mol L^{-1} s^{-1}. The time in which the concentration of reactants changed from 0.04 mol L^{-1} to 0.03 mol L^{-1} is

(1) 0.2 s

(2) 2.0 s

(3) 0.8 s

(4) 1.3 s

64. CH_3CH_2Br —KCN→ A (Major); —AgCN→ B (Major)

A and B are

(1) Optical isomers

(2) Functional isomers

(3) Position isomers

(4) Chain isomers

65. Correct order of boiling point of the given molecules is

(1) $(CH_3)_3C-Cl > (CH_3)_3C-Br > (CH_3)_3C-I$

(2) $(CH_3)_3C-Cl > (CH_3)_3C-I > (CH_3)_3C-Br$

(3) $(CH_3)_3C-I > (CH_3)_3C-Cl > (CH_3)_3C-Br$

(4) $(CH_3)_3C-I > (CH_3)_3C-Br > (CH_3)_3C-Cl$

66. Given below are two statements.
Statement (I): $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ is a displacement reaction.
Statement (II): Average oxidation number of bromine in tribromooctaoxide is $\frac{16}{3}$.
In the light of above statements choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

67. The standard Gibbs free energy change for the given cell reaction (in kJ mol^{-1}) will be
$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$
$(E^\circ_{Zn^{2+}/Zn} = -0.76\,V;\ E^\circ_{Ag^+/Ag} = 0.80\,V)$

(1) – 25.35

(2) – 150.21

(3) – 301.08

(4) – 501.12

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85. Match List-I with List-II for given balanced chemical reaction
$a\,MnO_4^- + b\,C_2O_4^{2-} + c\,H^+ \rightarrow a\,Mn^{2+} + d\,CO_2 + e\,H_2O$

List-I	List-II
(a) a	(i) 5
(b) b	(ii) 2
(c) c	(iii) 16
(d) d	(iv) 10

Correct match is

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

86. Henry's law constant (K_H) for the following gases, He, N_2 and O_2 are 144.97, 76.48 and 34.86 kbar respectively at 293 K. The correct order of solubility is

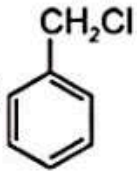
(1) $N_2 > O_2 > He$

(2) $He > O_2 > N_2$

(3) $O_2 > N_2 > He$

(4) $He > N_2 > O_2$

87. Which of the following is least likely to undergo S_N1 reaction with OH^- (aq)?

(1) 

(2) $CH_2 = CH - CHCl - CH_3$

(3) $(CH_3)_3CCl$

(4) $CH_3 - CH = CH - Cl$

88. pH of a saturated diacidic base $B(OH)_2$ is 10. The solubility product of $B(OH)_2$ at 25°C is

(1) 2×10^{-12}

(2) 5×10^{-13}

(3) 5×10^{-11}

(4) 2×10^{-16}

89. If for the reactions
$A \rightleftharpoons B$, $K_C = 10$,
$B \rightleftharpoons C$, $K_C = 5$ and
$C \rightleftharpoons D$, $K_C = 2$
then K_C for the reaction $A \rightleftharpoons D$ is

(1) 10

(2) 20

(3) 50

(4) 100

90. Given below are two statements, one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): On electrolysis of aqueous NaCl solution using inert electrodes, pH of the solution increases.
Reason (R) : On electrolysis of aqueous NaCl, water gets reduced at cathode and concentration of OH^- ion increases in solution.
In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

Vivek Krishna
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BOTANY

91. Which of the following sequences is **correct** for the development of embryo sac?

(1) Megaspore (2n) → Megaspore mother cell (n) → Embryo sac (n)

(2) Megaspore mother cell (n) → Megaspore (n) → Embryo sac (n)

(3) Nucellus (2n) → Megaspore mother cell (n) → Megaspore (n) →Embryo sac (n)

(4) Nucellus (2n) → Megaspore mother cell (2n) → Megaspore (n) → Embryo sac (n)

92. Identify the statements which are true for flowers pollinated by most common abiotic pollination agent
I. They possess well exposed stamens
II. Pollen grains are light and non-sticky
III. They have large coloured flowers
IV. They possess feathery stigma to easily trap air borne pollen grains
V. Numerous flowers packed into an inflorescence

(1) Only I and II

(2) Only III and IV

(3) II, III and IV

(4) I, II, IV and V

93. Filiform apparatus is present in the

(1) Largest cell of embryo sac

(2) Egg cell

(3) Synergid cell

(4) Antipodal cell

94. Egg apparatus in angiosperms is

(1) 7 celled 8 nucleated

(2) 8 celled 7 nucleated

(3) 3 celled 3 nucleated

(4) 3 celled 8 nucleated

95. Match the following columns w.r.t. pollen grains and select the **correct** option.

	Column-I		Column-II
a.	Plasma membrane	(i)	Thin and continuous inner wall
b.	Intine	(ii)	Lack outer hard covering
c.	Sporopollenin	(iii)	Made up of phospholipids
d.	Germ pore	(iv)	No known enzyme can degrade it

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iv), b(i), c(iii), d(ii)

96. Read the following statements and select the **correct** option.
Statement A: Pollen grains lose viability within 30 minutes of their release in members of Rosaceae, Leguminoseae and Solanaceae.
Statement B: Pollen grains can be stored in liquid nitrogen at –196° C for years.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

97. Select the **correct** sequence of anther wall layers from outermost to innermost

(1) Epidermis → Middle layer → Tapetum → Endothecium

(2) Epidermis → Endothecium → Middle layer → Tapetum

(3) Tapetum → Epidermis → Middle layer → Endothecium

(4) Tapetum → Middle layer → Epidermis → Endothecium

98. Match the columns and select the **correct** option.

Column I		Column II	
a.	Rhizoids	(i)	Tallest gymnosperm
b.	Hold fast	(ii)	Root like structure arise from thallus of bryophytes
c.	*Sequoia*	(iii)	Symbiosis with cyanobacteria
d.	Coralloid roots	(iv)	For attachment to the substratum

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(i), b(iii), c(iv), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

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99. Read the following statements and select the **correct** option.
Statement A : *Cycas* has branched stem while *Pinus* has unbranched stem.
Statement B : In *Cycas*, male cones and megasporophylls are borne on different plants.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

100. Artificial system of classification given by Linnaeus

(1) Was a phylogenetic classification system

(2) Was based upon both external and internal features

(3) Gave equal weightage to vegetative and sexual characteristics

(4) Was based upon the natural affinities among the organisms

101. Select the **wrongly** matched pair.

(1) *Ectocarpus* – I aterally attached flagella in zoospores

(2) *Sphagnum* – Zygotic meiosis absent

(3) *Funaria* – Vegetative reproduction by fragmentation

(4) *Porphyra* – Motile asexual spores

102. Cell wall is mainly made up of inner layer of cellulose and outer layer of pectose in

(1) *Ectocarpus*

(2) *Dictyota*

(3) *Chara*

(4) *Polysiphonia*

103. In mosses, juvenile stage
a. Is part of gametophytic phase
b. Arises from zygote
c. Reproduces vegetatively
d. Gives rise to mature leafy gametophyte
Choose the **correct** ones.

(1) a and b only

(2) b, c and d

(3) a and c only

(4) a, c and d

104. The most crucial step of Calvin cycle is

(1) Carboxylation

(2) Oxidation

(3) Regeneration

(4) Reduction

105. During C_4 pathway, decarboxylation of C_4 acid and regeneration of PEP respectively occur in

(1) Bundle sheath cell and epidermal cell

(2) Bundle sheath cell and mesophyll cell

(3) Mesophyll cell and bundle sheath cell

(4) Epidermal cell and mesophyll cell

106. In non-cyclic photophosphorylation

(1) Only ATP is synthesised

(2) NADPH is converted into $NADP^+$

(3) Both ATP and NADPH are formed

(4) Oxygen is not evolved

107. In C_4 plants, C_4 acid from mesophyll is broken down in the bundle sheath cells to release CO_2, which results in increase in intracellular concentration of CO_2. This ensures that RuBisCO functions as

(1) Oxygenase and minimises carboxylase activity

(2) Carboxylase and minimises its oxygenase activity

(3) Carboxylase as well as oxygenase

(4) Oxygenase and maximises its carboxylase activity

108. The reaction centre w.r.t. cyclic photophosphorylation is

(1) P_{700}

(2) P_{680}

(3) P_{540}

(4) P_{660}

109. Arithmetic growth can be expressed by which one of the following equations?

(1) $W_1 = W_0 e^{rt}$

(2) $L_t = L_0 + rt$

(3) $W_0 = W_1 e^{rt}$

(4) $L_0 = L_t + e^{rt}$

110. The plant in which the shape of leaves in terrestrial habitat differ from that of aquatic habitat is

(1) Coriander

(2) Butter cup

(3) Cotton

(4) Larkspur

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111. To promote flowering in pineapples, which of the following PGRs can be used?

(1) Gibberellins

(2) Cytokinins

(3) Auxin

(4) ABA

112. The plant hormone used in improvement of shape of apple fruit and can speed up malting process in brewing industry is

(1) Ethylene

(2) Auxins

(3) Gibberellins

(4) Cytokinins

113. How many molecules of NADH + H^+ and FADH + H^+ are respectively produced during the complete oxidation of one molecule of pyruvic acid in mitochondria?

(1) 3 and 1

(2) 2 and 2

(3) 8 and 2

(4) 4 and 1

114. Select the **incorrect** statement w.r.t TCA cycle in eukaryotes.

(1) The cycle starts with the condensation of acetyl group with OAA and water

(2) TCA cycle is confined to mitochondrial matrix

(3) This cycle serves as a pathway for oxidation of carbohydrates only

(4) Succinate dehydrogenase involved in this process is found attached to inner mitochondrial membrane

115. During electron transport system in mitochondria, which of the following complex contain cytochromes a and a_3 and two copper centers?

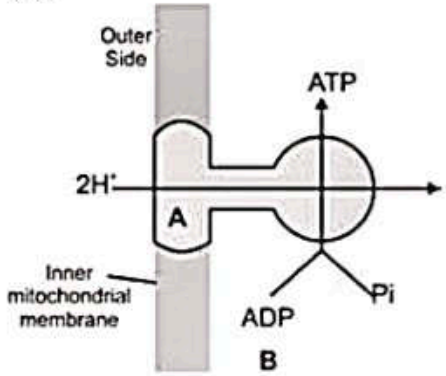
(1) Complex I

(2) Complex II

(3) Complex IV

(4) Complex V

116. Consider the given figure and identify the part labelled as **A** & **B**.



(1) **A** is a peripheral membrane protein

(2) **B** is intermembrane space of mitochondria

(3) **B** is the site of TCA cycle

(4) **A** is headpiece of an integral protein

117. The mobile electron carrier attached to outer surface of inner membrane of mitochondria is

(1) Cytochrome *a*

(2) Ubiquinone

(3) Cytochrome a_3

(4) Cytochrome *c*

118. Conversion of glucose to glucose-6-phosphate is catalysed by

(1) Enolase

(2) Hexokinase

(3) Pyruvate kinase

(4) PFK

119. Mark true (T) or false (F) for the following statements and choose the **correct** option.

(a) In a single turn of citric acid cycle, CO_2 is released at three steps.

(b) NADH + H^+ is produced in three steps of citric acid cycle.

(c) Conversion of succinic acid into malic acid leads to formation of $FADH_2$ in Krebs' cycle.

(d) Succinic acid is the four carbon molecule produced during the conversion step where substrate level phosphorylation occurs.

(1) (a)-(F), (b)-(T), (c)-(T), (d)-(F)

(2) (a)-(F), (b)-(F), (c)-(T), (d)-(T)

(3) (a)-(T), (b)-(T), (c)-(F), (d)-(F)

(4) (a)-(F), (b)-(T), (c)-(T), (d)-(T)

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120. Water containing cavities are seen inside vascular bundles of

(1) Dicot root

(2) Monocot stem

(3) Dicot stem

(4) Monocot root

121. Read the following **assertion** (A) and **reason** (R) and select the **correct** option

A : Trichomes are branched unicellular appendages arising from root.

R : Trichomes facilitates water loss by transpiration.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

122. Ground tissue of leaves is called

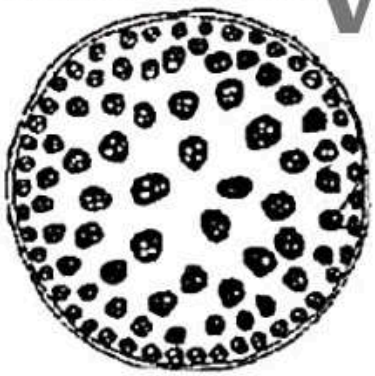
(1) Conjunctive tissue

(2) Stelar tissue

(3) Mesophyll tissue

(4) Conducting tissue

123. The figure depicted below is transverse section of



(1) Monocot stem

(2) Monocot root

(3) Dicot stem

(4) Dicot root

124. Which of the following **cannot** be floral reward provided by plants to pollinators?

(1) Pollens

(2) Nectar

(3) Fragrance

(4) Providing safe place to lay eggs

125. Read the following assertion (A) and reason (R) and select the **correct** option.

Assertion (A): Ethylene increases the yield of cucumbers.

Reason (R): Ethylene promotes female flowers in cucumbers.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

126. The thallus is dorsiventral and closely appressed to the substratum in

(1) *Funaria*

(2) *Sphagnum*

(3) *Marchantia*

(4) *Polytrichum*

127. The respiratory quotient of $C_{51}H_{98}O_6$ (tripalmitin) is

(1) 1.4

(2) 1

(3) 0.9

(4) 0.7

128. Find the **incorrect** statement w.r.t. fermentation.

(1) Pyruvic acid is converted to $CO_2 + H_2O$ + Energy.

(2) Occurs in bacteria such as LAB

(3) Either acid or alcohol is produced during the process

(4) The reducing agent is NADH + H^+ which is reoxidised to NAD^+

129. The most abundant enzyme in the world

a. Is characterised by the fact that its active site can bind to both CO_2 and O_2.

b. Has a much greater affinity for CO_2 when the $CO_2 : O_2$ is nearly equal.

c. Is present in the mesophyll cells of the C_4 plants.

d. Is responsible for fixing CO_2 with 3-carbon molecule phosphoenol pyruvate in C_4 plants.

The **correct** ones are

(1) Only a and d

(2) Only b and c

(3) Only a and b

(4) Only b and d

130. Choose the **odd** one w.r.t. members of class Pteropsida.

(1) *Dryopteris*

(2) *Adiantum*

(3) *Lycopodium*

(4) *Pteris*

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ZOOLOGY

136. What happens to GFR due to constrictory effect of ADH on peripheral blood vessels?

(1) Increases

(2) Decreases

(3) Tends to remain at the same level

(4) First increases then decreases

137. Other than kidneys, how many of the following organ(s) also help in the elimination of excretory wastes in humans?

Liver, Skin, Lungs, Sebaceous gland, Salivary gland

Select the **correct** option.

(1) Four

(2) Five

(3) Two

(4) One

138. The basic functional unit of human kidney is

(1) Nephron

(2) Renal pyramid

(3) Nephridia

(4) Henle's loop

139. The filtration slits are formed by which of the following cells/layers of filtration membrane?

(1) Endothelium of glomerular blood vessels

(2) Podocytes of Bowman's capsule

(3) Endothelium of vasa recta

(4) Basement membrane of PCT

140. The medullary interstitial fluid has a very high osmolarity which is maintained by

(1) Movement of urea from collecting duct into medullary interstitium

(2) Movement of NaCl from descending limb of Henle's loop into medullary interstitium

(3) Movement of K^+ and Cl^- from PCT and DCT into peri-tubular capillaries

(4) Uric acid and H^+ present in interstitium

141. In a healthy adult human, urine is

(1) Hypotonic to blood

(2) Hypertonic to blood

(3) Isotonic to blood

(4) Hypo-osmotic to blood

142. In humans, cortical nephrons differ from juxta-medullary nephrons in all of the following aspects, **except**

(1) Number

(2) Length of loop of Henle

(3) Length of vasa recta

(4) Presence of peritubular capillaries

143. Part of nephron lined by simple cuboidal brush bordered epithelium is

(1) Loop of Henle

(2) Glomerulus

(3) Proximal convoluted tubule

(4) Collecting duct

144. Match column I with column II

	Column I		Column II
a.	Glycosuria	(i)	Inflammation of glomeruli
b.	Gout	(ii)	Mass of crystallised salts within the kidney.
c.	Renal calculi	(iii)	Presence of glucose in urine
d.	Glomerulo-nephritis	(iv)	Accumulation of uric acid crystals in joints

Choose the correct option

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iii), b(ii), c(iv), d(i)

145. Each rib has two articulation surfaces on its dorsal end and is hence called

(1) Bicephalic

(2) Vertebrochondral

(3) Vertebrosternal

(4) Vertebral

146. Read the statements carefully and choose the **correct** option.

Statement A: Actin and myosin are polymerised contractile proteins.

Statement B: Motor neurons carry signals to the muscle fibres which generate an action potential.

(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

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147. Select the **mismatch** w.r.t joints in humans.

(1) Saddle joint – Between carpal and metacarpal of thumb.

(2) Pivot joint – Between atlas and axis

(3) Hinge joint – Between humerus and ulna

(4) Gliding joint – Between phalanges

148. The type of muscle fibres present in our

(1) Biceps are uninucleated and voluntary

(2) Heart are non-striated and involuntary

(3) Thigh are multinucleated and fusiform in shape

(4) Stomach are unbranched and involuntary

149. In the human body, the total number of floating ribs and collar bones respectively are

(1) 4 and 2

(2) 6 and 4

(3) 14 and 4

(4) 4 and 8

150. Number of glenoid cavity(ies) associated with pectoral girdle is/are

(1) One

(2) Three

(3) Two

(4) Zero

151. Acromion process is related to which bone?

(1) Clavicle

(2) Scapula

(3) Radius

(4) Ulna

152. Chose the **odd** one from the following w.r.t. paired bones of skull.

(1) Lacrimal bone

(2) Ethmoid bone

(3) Zygomatic bone

(4) Palatine bone

153. How many of the statements mentioned below are **incorrect**?

A. Head of humerus articulates with glenoid cavity present above acromion process

B. Patella covers the knee dorsally

C. Clavicle also articulates with sternum

D. Fibula is present on medial aspect of hind limb

Select the **correct** option.

(1) 1

(2) 2

(3) 3

(4) 4

154. Which one is responsible for maintaining body temperature?

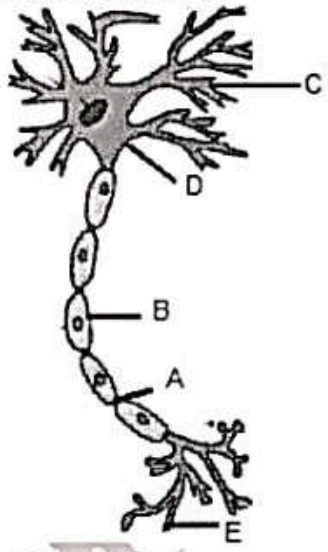
(1) Medulla oblongata

(2) Cerebellum

(3) Cerebrum

(4) Hypothalamus

155. In the given figure of neuron, which of the labelled parts have Nissl's granules?



(1) A, B & C

(2) A, C & E

(3) C & D only

(4) A, B, C & D

156. Type of neurons which comprise of one axon only and are found usually in embryonic stage are termed as

(1) Pseudounipolar

(2) Unipolar

(3) Bipolar

(4) Multipolar

157. Read the following statements

Statement-A : Diffusion of K^+ from axoplasm to ECF restores the resting membrane potential at the site of excitation of the nerve fibre.

Statement-B : Upon stimulation at a site on the polarised membrane, rapid influx of Na^+ occurs because the membrane becomes freely permeable to Na^+.

Choose the **correct** option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are incorrect

(4) Both statements are correct

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158. Meninx which is in contact with the brain tissue and present in the form of a highly vascular thin layer is

(1) Dura mater

(2) Arachnoid

(3) Pia mater

(4) Coracoid

159. Which one is **not** a part of brain stem?

(1) Cerebrum

(2) Midbrain

(3) Pons

(4) Medulla oblongata

160. The part of the human brain that consists of fibre tracts which interconnect different regions of the brain and contains a centre that can moderate the functions of the respiratory rhythm centre is

(1) Medulla oblongata

(2) Cerebellum

(3) Pons

(4) Hypothalamus

161. Select the **incorrect** statement about electrical synapses

(1) They are rare in our system

(2) Impulse transmission is faster in them as compared to a chemical synapse

(3) Pre and post synaptic membranes possess a wide gap between them

(4) Neurotransmitters are absent at these synapses

162. How many hormones given in the box below are peptide hormones?

Erythropoietin, Estradiol, Glucagon, Thymosin, Parathyroid hormone, Adrenaline

Choose the **correct** option.

(1) Six

(2) Five

(3) Four

(4) Three

163. The hormone synthesised by the hypothalamus and released into the hypophyseal portal circulation to ensure production of certain hormones by the pituitary gland is

(1) ADH

(2) LH

(3) GnRH

(4) FSH

164. Hormones are

(1) Non-nutrient chemicals that act as intercellular messengers

(2) Nutrient chemicals that act as intercellular messengers

(3) Secretory products that act as intracellular messengers

(4) Volatile chemicals necessary for all body activities

165. Hormone used in transplantation surgery to suppress the immune response in the body of recipients are

(1) Mineralocorticoid

(2) Thyroxine

(3) Glucocorticoid

(4) Sex corticoid

166. A hormone that must generate secondary messenger for its action is the one that

(1) Maintains and supports pregnancy due to its steroidal nature

(2) Is the main mineralocorticoid of the body

(3) Maintains basal metabolic rate of body

(4) Stimulates the secretion of androgens from testes

167. Thymus gland is a lobular structure located between lungs behind _____ on the _____ side of aorta. Choose the option which **correctly** fills the blank respectively.

(1) Clavicle, dorsal

(2) Clavicle, lateral

(3) Sternum, ventral

(4) Sternum, dorsal

168. Endocrine gland that secretes melatonin and is believed to be involved in establishing biorhythms is

(1) Pituitary

(2) Pineal

(3) Thyroid

(4) Thymus

169. In human females, secondary sexual characters are controlled by

(1) Progesterone

(2) Parathormone

(3) Estrogen

(4) FSH

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170. **Assertion (A):** Androgens act on the central nervous system and influence the male libido.
Reason (R): Androgens play a major stimulatory role in the process of spermatogenesis.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

171. A complete alimentary canal with well-developed muscular pharynx is present in all the following, **except**

(1) *Ancylostoma*

(2) *Taenia*

(3) *Ascaris*

(4) *Pheretima*

172. An animal having lateral unsegmented appendages, dioecious with both external and internal metameric segmentation and indirect development is

(1) *Culex*

(2) Prawn

(3) Scorpion

(4) *Nereis*

173. _____ is a coelomate having open type of circulatory system.
Select the option which fills the blank **correctly.**

(1) Silkworm

(2) Roundworm

(3) Hookworm

(4) Earthworm

174. Select the feature that is **incorrect** for cyclostomes.

(1) Endoskeleton is cartilaginous

(2) Suctorial mouth and rasping jaws

(3) 6 – 15 pairs of gill slits

(4) Presence of closed circulatory system

175. All the given features (a-d) are present in
a. Sexes are separate
b. Fertilisation is external
c. Oviparity
d. Indirect development

(1) *Rana*

(2) *Balaenoptera*

(3) *Pheretima*

(4) *Macaca*

176. In the heart of a frog, a chamber that pumps out mixed blood, opens into _____ which is present on the ventral side of the heart.
Select the **correct** option to fill in the blank.

(1) Conus arteriosus

(2) Sinus venosus

(3) Left atrium

(4) Right atrium

177. Choose the **incorrect** statement w.r.t blood vascular system in *Periplaneta*.

(1) Haemolymph is composed of colourless plasma and haemocytes

(2) Heart is 13-chambered and lies along mid dorsal line of thorax and abdomen

(3) Blood from sinuses enter heart through ostia and is pumped posteriorly due to action of 12 alary muscles

(4) Visceral organs located in the haemocoel are bathed in haemolymph

178. Mushroom-shaped gland which is a characteristic of male cockroach is present in

(1) 6th - 7th abdominal segments

(2) 2nd - 6th abdominal segments

(3) 2nd and 3rd thoracic segments

(4) 9th - 10th abdominal segments

179. Copulatory pad in frog is present on the first digit of

(1) Forelimbs in males

(2) Hindlimbs in males

(3) Forelimbs in females

(4) Hindlimbs in females

180. Which of the following features **cannot** be associated with *Rana tigrina*?

(1) Tympanum represents the ear

(2) Three chambered heart

(3) Body divisible into head and trunk

(4) Direct development

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